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EX: 6 EXPONENTIAL SMOOTHING METHODS

AIM

To implement and compare three exponential smoothing techniques for time-series forecasting: Simple Exponential Smoothing, Holt's Linear Trend Method, and Holt-Winters Seasonal Method.

ALGORITHM

1. Start the program.
2. Import Pandas, Matplotlib, and the required exponential smoothing models.
3. Apply Simple Exponential Smoothing to sales data without a clear trend or seasonal pattern.
4. Forecast the next three values.
5. Apply Holt's method to sales data containing an increasing trend.
6. Forecast the next three values using Holt's method.
7. Apply Holt-Winters to seasonal sales data using additive trend and seasonality.
8. Forecast the next four values using Holt-Winters.
9. Plot the actual and fitted values for all three methods.
10. Compare the results and stop the program.

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PROGRAM

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.holtwinters import (
    SimpleExpSmoothing, Holt, ExponentialSmoothing
)

sales = [100,120,110,115,105,125,118,122,110,119]
data = pd.Series(sales)

model1 = SimpleExpSmoothing(
    data, initialization_method="estimated"
)
fit1 = model1.fit(smoothing_level=0.3, optimized=False)
forecast1 = fit1.forecast(3)

sales_trend = [
    100,105,112,118,125,
    130,138,145,152,160
]
data_trend = pd.Series(sales_trend)

model2 = Holt(
    data_trend, initialization_method="estimated"
)
fit2 = model2.fit()
forecast2 = fit2.forecast(3)

seasonal_sales = [
    100,130,160,120,110,145,175,130,
    120,155,190,140,130,165,205,150
]
data_seasonal = pd.Series(seasonal_sales)

model3 = ExponentialSmoothing(
    data_seasonal,
    trend="add",
    seasonal="add",
    seasonal_periods=4,
    initialization_method="estimated"
)
fit3 = model3.fit()
forecast3 = fit3.forecast(4)

print("Simple Exponential Smoothing")
print(forecast1)
print("\nHolt's Method")
print(forecast2)
print("\nHolt-Winters Method")
print(forecast3)

plt.plot(data, marker='o', label='Actual')
plt.plot(fit1.fittedvalues, marker='o', label='Forecast')
plt.title("Simple Exponential Smoothing")
plt.xlabel("Time")
plt.ylabel("Sales")
plt.legend()
plt.grid(True)
plt.show()

plt.plot(data_trend, marker='o', label='Actual')
plt.plot(fit2.fittedvalues, marker='o', label='Holt Forecast')
plt.title("Holt's Method")
plt.xlabel("Time")
plt.ylabel("Sales")
plt.legend()
plt.grid(True)
plt.show()

plt.plot(data_seasonal, marker='o', label='Actual')
plt.plot(fit3.fittedvalues, marker='o',
         label='Holt-Winters Forecast')
plt.title("Holt-Winters Method")
plt.xlabel("Quarter")
plt.ylabel("Sales")
plt.legend()
plt.grid(True)
plt.show()
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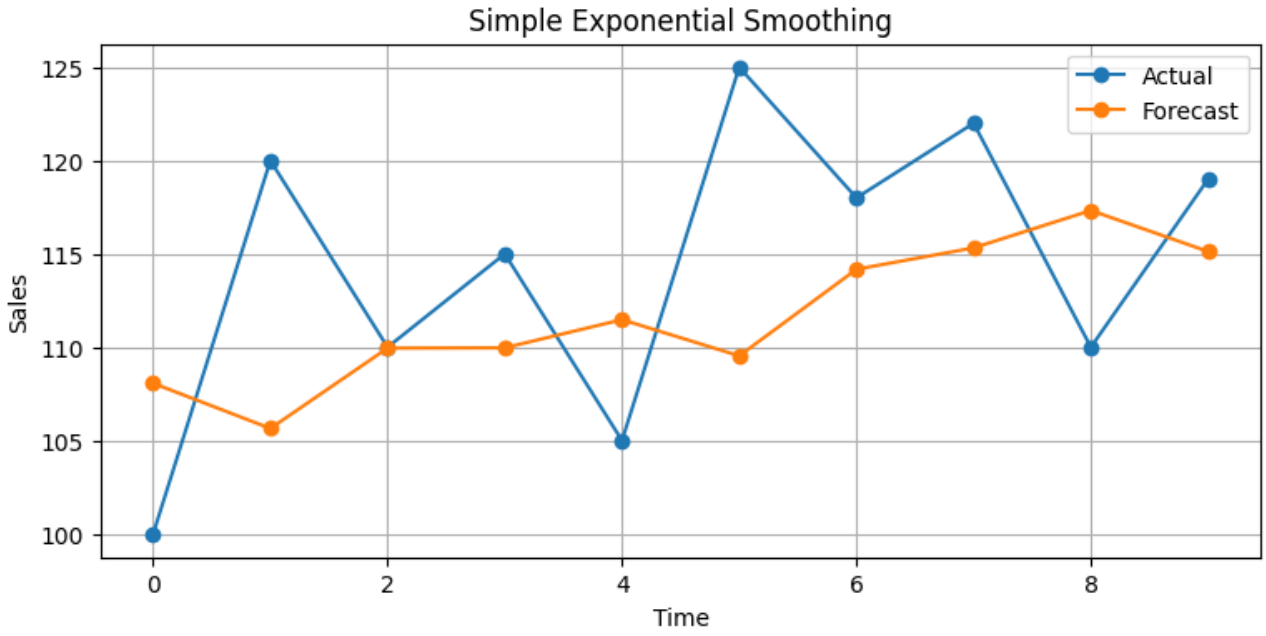
OUTPUT

```
Simple Exponential Smoothing
10    116.28929
11    116.28929
12    116.28929
dtype: float64

Holt's Method
10    165.133284
11    171.793879
12    178.454475
dtype: float64

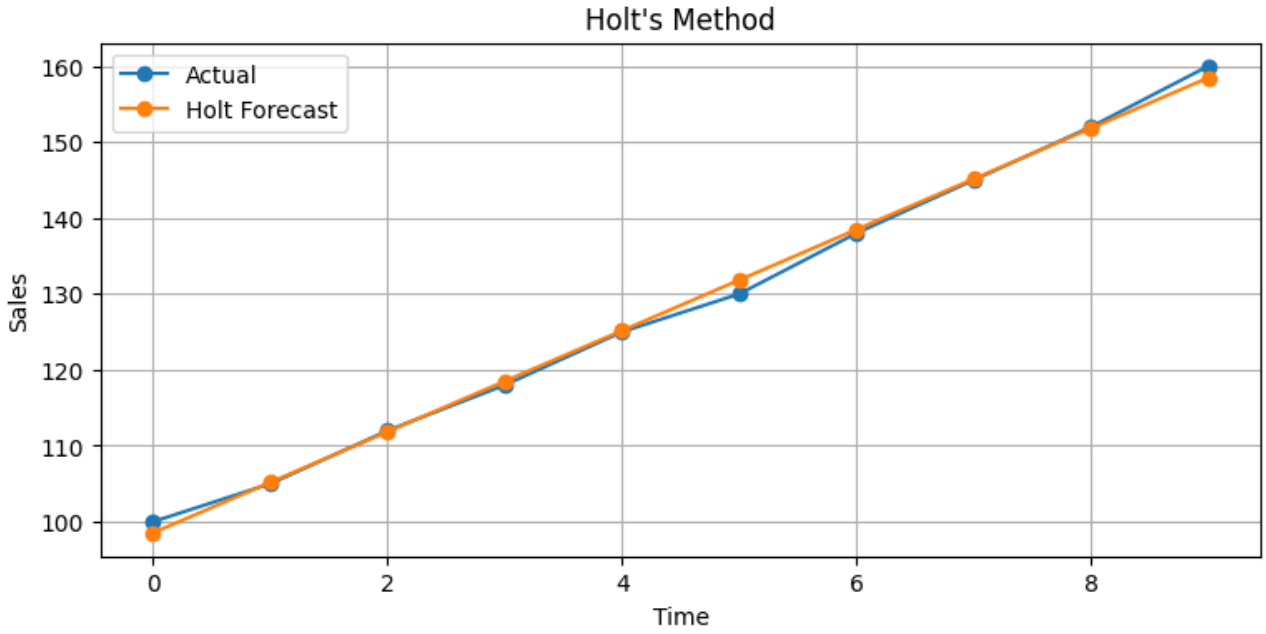
Holt-Winters Method
16    144.062516
17    177.812488
18    211.562504
19    164.062511
dtype: float64
```

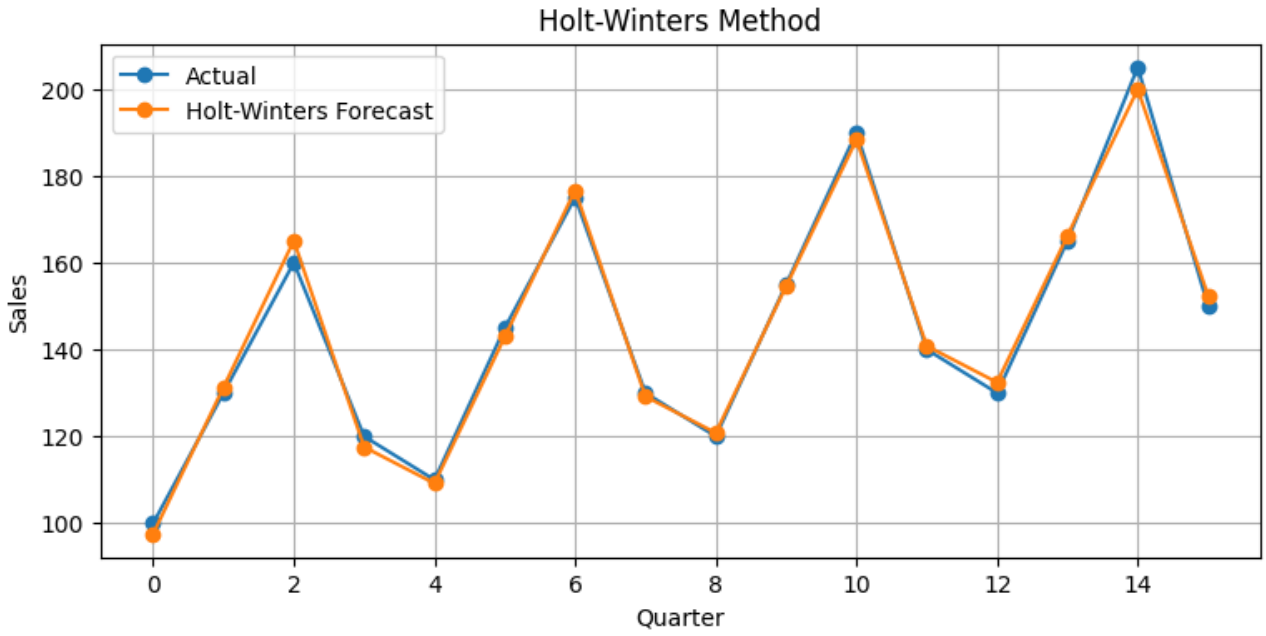
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GRAPHS


Holt's Method
Actual
Holt Forecast
Sales
Time
160
150
140
130
120
110
100
0
2
4
6
8


Holt-Winters Method
Actual
Holt-Winters Forecast
Sales
Quarter
200
180
160
140
120
100
0
2
4
6
8
10
12
14

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OBSERVATION

Simple Exponential Smoothing follows the level of the data, Holt's method follows the increasing trend, and Holt-Winters captures both the trend and the repeating seasonal pattern.

RESULT

Thus, Simple Exponential Smoothing, Holt's Method, and Holt-Winters Method were successfully implemented and compared for time-series forecasting.